Time Speed & Distance

Level -1 Foundation

1. A man travels first 50 km at 25 kmph, next 40 km at 20 kmph and then 90 km at 15 kmph. His average speed for the whole journey (in kmph) is :
   (a) 25
   (b) 20
   (c) 18
   (d) 40
   (e) None of these
2. A man walks at the rate of 5 km/hr for 6 hours and at 4 km/hr for 12 hours. The average speed of the man (in km/hr) is :
   (a) 4
   (b) 4.5
   (c) 4.33
   (d) 4.66
   (e) None of these
3. If a person travels 51/5 km in 3 hours, then the distance covered by him in 5 hours will be :
   (a) 18 km
   (b) 17 km
   (c) 16 km
   (d) 15 km
   (e) None of these
4. If a train 110m long passes a telegraph pole in 3 seconds, then the time taken (in seconds) by it to cross a railway platform 165 m long, is :
   (a) 3
   (b) 4
   (c) 5
   (d) 7.5
   (e) None of these
5. A train 700 m long is running at the speed of 72 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in metres) is :
   (a) 700
   (b) 600
   (c) 550
   (d) 500
   (e) None of these
6. If a 200 m long train crosses a platform of the same length as that of the train in 20 seconds, then the speed of the train is :
   (a) 50 km/hr
   (b) 60 km/hr
   (c) 72 km/hr
   (d) 80 km/hr
   (e) None of these
7. Two trains, each of length 125 metre, are running in parallel tracks in opposite directions. One train is running at a speed 65 km/hour and they cross each other in 6 seconds. The speed of the other train is:
   (a) 75 km /hour
   (b) 85 km/ hour
   (c) 95 km/ hour
   (d) 105 km/ hour
   (e) None of these

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8. A man with 3/5 of his usual speed reaches the destination 2.5 hours late. Find his usual time to reach the destination?
   (a) 4 hours
   (b) 3 hours
   (c) 3.75 hours
   (d) 4.5 hours
   (e) None of these
9. A train running at 7/11 of its normal speed reached a place in 22 hours. How much time could be saved if the train would have run at its normal speed?
   (a) 14 hours
   (b) 7 hours
   (c) 8 hours
   (d) 16 hours
   (e) None of these
10. Walking at three-fourth of his usual speed, a man covers certain distance in 2 hours more than the time he takes to cover the distance at his usual speed. The time taken by him to cover the distance with his usual speed is:
   (a) 4.5 hours
   (b) 5.5 hours
   (c) 6 hours
   (d) 5 hours
   (e) None of these
11. A man goes from a place A to B at a speed of 12 km/hr and returns from B to A at a speed of 18 km/hr. The average speed for the whole journey is:
   (a) 72/5 km/hr
   (b) 15 km/hr
   (c) 31/2 km/hr
   (d) 16 km/hr
   (e) None of these
12. Two trains started at the same time, one from A to B and the other from B to A. If they arrived at B and A respectively 4 hours and 9 hours after they passed each other, the ratio of the speeds of the two trains was:
   (a) 2 : 1
   (b) 3 : 2
   (c) 4 : 3
   (d) 5 : 4
   (e) None of these
13. A starts from a place P to go to a place Q. At the same time B starts from Q to P. If after meeting each other A and B took 16 and 25 hours more respectively to reach their destinations, the ratio of their speeds is:
   (a) 3 : 2
   (b) 5 : 4
   (c) 9 : 4
   (d) 9 : 13
   (e) None of these
14. A train of 320 m cross a platform in 24 seconds at the speed of 120 km/ h. while a man cross same platform in 4 minute. What is the speed of man in m/s?
   (a) 2.4
   (b) 1.5
   (c) 1.6
   (d) 2.0
   (e) None of these
15. A car travel first 39 km distance in 45 minute while next 25 km distance in 35 minutes. What is its average speed?

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(a) 45 Km/h
(b) 35 Km/h
(c) 48 Km/h
(d) 90 Km/h
(e) None of these

16. A 280 m long train moving with an average speed of 108 km/h cross a platform in 12 seconds. A boy crosses the same platform in 10 seconds. What is the speed of boy in m/s?
(a) 5 m/s
(b) 8 m/s
(c) 10 m/s
(d) Cannot be determined
(e) None of these

17. A truck cover 224 km in 4 hours, the average speed of a bike is 1/4th the average speed of the truck how much distance will the bike cover in seven hour?
(a) 96 km
(b) 98 km
(c) 95 km
(d) 92 km
(e) None of these

18. Train fare between Patna to Munger for one adult is three times the train fair of one child. If adult fair is 102 then. What will be the fare of 2 adult and 3 children together for same distance?
(a) 306
(b) 212
(c) 206
(d) 214
(e) None of these

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19. A car travels a distance of 75 km at the speed of 25 km/h. It covers the next 25 km of its journey at the speed of 5 km/h and the last 50 km of its journey at the speed of 25 km/h. What is the average speed of car?
(a) 15 Km/h
(b) 12.5 Km/h
(c) 40 km/h
(d) 25 km/h
(e) None of these

20. If the length of the train is 700 m and length of platform is 500 m. Find the time taken by the train moving at 54 km/h to cross platform.
(a) 75 sec
(b) 80 sec
(c) 85 sec
(d) 90 sec
(e) None of these

21. Amit start to go to Delhi from patna at speed of 50 km/h. Distance between Delhi and Patna is 1000 km. He takes rest of 20 minutes every 3 hours of journey. How much time will he take to arrive Delhi?
(a) 20 hours
(b) 21 hours
(c) 22 hours
(d) 23 hours
(e) None of these

22. A car cover a distance of 330 km in a certain amount of time at speed of 55 km/h. What is the average speed of bike that cover distance of 15 km less than car in 1 hour less than time taken by car?
(a) 50 Km/h
(b) 60 Km/h

(c) 63 Km/h
(d) 65 Km/h
(e) None of these

23. A car traveling at a speed of 40 km/hour can complete a journey in 9 hours. How long will it take to travel the same distance at 60 km/ hour?
(a) 6 hours
(b) 3 hours
(c) 4 hours
(d) 17/4 hours
(e) None of these

24. A 75 meter long train is moving at 20 kmph. It will cross a man standing on the platform in
(a) 12 seconds
(b) 14 seconds
(c) 13.5 seconds
(d) 15.5 seconds
(e) None of these

25. In what time will a train 100 meter long cross an electric pole, if its speed be 144 km/hour?
(a) 2.5 seconds
(b) 5 seconds
(c) 12.5 seconds
(d) 3 seconds
(e) None of these

26. A train running at a speed of 60 kmph crosses a platform double its length in 32.4 seconds. What is the length of the platform?
(a) 180 m
(b) 240 m
(c) 360 m
(d) 90 m
(e) None of these

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27. A train running at the speed of 66 kmph crosses a signal pole in 18 seconds. What is the length of the train?
(a) 330 m
(b) 300 m
(c) 360 m
(d) 320 m
(e) None of these

28. Two trains of equal length take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 120 metres, in what time (in seconds) will they cross each other travelling in opposite direction?
(a) 16 sec
(b) 15 sec
(c) 12 sec
(d) 10 sec
(e) None of these

29. A man covers a certain distance by car driving at 70 km/hr and he returns to the starting point riding on a scooter at 55 km/hr. Find his average speed for the whole journey.
(a) 61.6 km/hr
(b) 62.8 km/hr
(c) 63.6 km/hr
(d) 64.6 km/hr
(e) None of these

30. A boy walking at a speed of 10 km/hr reaches his school 15 minutes late. Next time he increases his speed by 2 km/hr, but still he is late by 5 minutes. Find the distance of his school from his house.

(a) 10 km
(b) 15 km
(c) 20 km
(d) 25 km
(e) None of these

31. A motor car does a journey in 10 hrs, the first half at 21 km/hr and the second half at 24 km/hr. Find the distance?
(a) 224 km
(b) 225 km
(c) 226 km
(d) 228 km
(e) None of these

32. Two men A and B starts from a place P walking at 3 km and 3.5 km an hour respectively. How many kms will they be apart at the end of 3 hrs, if they walk in opposite directions?
(a) 13.5 km
(b) 15.5 km
(c) 17.5 km
(d) 19.5 km
(e) None of these

33. Two men A and B starts from a place P walking at 3 km and 3.5 km an hour respectively. How many kms will they be apart at the end of 3 hrs, if they walk in opposite directions?
(a) 13.5 km
(b) 15.5 km
(c) 17.5 km
(d) 19.5 km
(e) None of these

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34. What is the length of the bridge on which a man riding 15 km an hour can cross in 5 minutes?
(a) 850 m
(b) 1050 m
(c) 1250 m
(d) Can't be determined
(e) None of these

35. A train moving with the speed of 180 km/hr. Its speed (in metres per second) is:
(a) 5
(b) 40
(c) 30
(d) 50
(e) None of these

36. If a train, with a speed of 60 km/hr, crossed a pole in 30 second, the length of the train (in metres) is:
(a) 1000
(b) 900
(c) 750
(d) 500
(e) None of these

37. Two trains are moving in the same direction at 50 km/hr and 30 km/hr. The faster train crossed a man in the slower train in 18 seconds. Find the length of the faster train?
(a) 80 m
(b) 90 m
(c) 100 m
(d) Can't be determined
(e) None of these

38. A train running at 25 km/hr takes 18 seconds to pass a platform. Next, it takes 12 seconds to pass a man walking at 5 km/hr in the opposite direction. Find the sum of the length of the train and that of the platform?
    (a) 125 m
    (b) 135 m
    (c) 145 m
    (d) 155 m
    (e) None of these
39. A train is travelling at a rate of 45 km/hr. How many seconds, it will take to cover a distance of 4/5 km?
    (a) 36 second
    (b) 64 second
    (c) 90 second
    (d) 120 second
    (e) None of these
40. A 570-metre-long train crosses a platform of equal length in 15 seconds. What is the speed of the train in metres/second?
    (a) 38
    (b) 54
    (c) 76
    (d) 70
    (e) None of these

Level -2 Moderate

1. A boy is running at a speed of p kmph to cover a distance of 1 km. But, due to the slippery ground, his speed is reduced by q kmph (p > q). If he takes r hours to cover the distance, then:
    (a) $1/r = 1/p + 1/q$
    (b) $1/r = p - q$
    (c) $r = p + q$
    (d) $r = p - q$
    (e) None of these
2. A boy goes to the school with the speed of 3 kmph and returns with the speed of 2 kmph. If he takes 5 hours in all, then the distance (in kms) between the village and the school is :
    (a) 6
    (b) 12
    (c) 8
    (d) 9
    (e) None of these
3. A student walks from his house at 5 kmph and reaches his school 10 minutes late. If his speed had been 6 kmph he would have reached 15 minutes early. The distance of his school from his house is :
    (a) 2.5 km
    (b) 12.5 km
    (c) 5.5 km
    (d) 3.6 km
    (e) None of these
4. A train takes 18 seconds to pass completely through a station 162 m long and 15 seconds through another station 120 m long. The length of the train (in metres) is :
    (a) 70
    (b) 80
    (c) 90
    (d) 100
    (e) None of these

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5. Two trains of lengths 120 m and 80 m are running in the same direction with velocities of 40 kmph and 50 kmph respectively. The time taken by them to cross each other is :
   (a) 60 sec
   (b) 72 sec
   (c) 75 sec
   (d) 80 sec
   (e) None of these
6. A man standing on a railway platform observes that a train going in one direction takes 4 seconds to pass him. Another train of same length going in the opposite direction takes 5 seconds to pass him. The time taken (in seconds) by the two trains to cross each other will be :
   (a) 31/9
   (b) 40/9
   (c) 49/9
   (d) 50/9
   (e) None of these
7. A train, 240m long crosses a man, walking along the line in opposite direction at the rate of 3 kmph in 10 seconds. The speed of the train is:
   (a) 63 km/h
   (b) 75 km/h
   (c) 83.4 km/h
   (d) 86.4 km/h
   (e) None of these
8. The distance between two cities A and B is 330km. A train starts from A at 8 am. and travels towards B at 60 km/hr. Another train starts from B at 9 am. and travels towards A at 75 km/hr. At what time do they meet?
   (a) 10 am.
   (b) 10 : 30 am.
   (c) 11 am.
   (d) 11 : 30 am.
   (e) None of these
9. Two men are standing on opposite ends of a bridge of 1200 metres long. If they walk towards each other at the rate of 5m/minute and 10m/minute respectively, in how much time will they meet each other? (a) 60 minute
   (b) 80 minute
   (c) 85 minute
   (d) 90 minute
   (e) None of these
10. Two trains one 160 m and the other 140 m long are running in opposite directions on parallel track, the first at 77 km an hour and the other at 67 km an hour. How long will they take to cross each other?
   (a) 7 seconds
   (b) 7.5 seconds
   (c) 6 seconds
   (d) 10 seconds
   (e) None of these
11. Two trains of equal length take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 120 meters, in what time (in seconds) will they cross each other travelling in same direction?
   (a) 16
   (b) 15
   (c) 12
   (d) 60
   (e) None of these

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12. A person travels 600 km by train at 80km/hr, 800 km by ship at 40 km/ hr. 500 km by aeroplane at 400 km/hr and 100 km by car at 50 km/hr. What is the average speed for the entire distance? (a) 5 65 123 km./hr.
    (b) 60 km./hr.
    (c) 5 60 123 km./hr.
    (d) 62 km./ hr.
    (e) None of these
13. One third of a certain journey is covered at the rate of 25 km/hr one fourth at the rate of 30 km/hr and the rest at 50 km/hr. The average speed for the whole journey is
    (a) 35 km/hr
    (b) 33.33 km/hr
    (c) 30 km/hr
    (d) 445/12 km/hr
    (e) None of these
14. In covering certain distance, the speeds of A and B are in the ratio of 3 : 4 A takes 30 minutes more than B to reach the destination. The time taken by A to reach the destination is :
    (a) 1 hour
    (b) 3/2 hours
    (c) 2 hours
    (d) 5/2 hours
    (e) None of these
15. A car starts with the speed of 70 km/h with its speed increasing every two hours by 10 km/h. In how many hours will it cover 345 km?
    (a) 9/4 hours
    (b) 4 hours
    (c) 9/2 hours
    (d) 3 hour
    (e) None of these
16. Aditya rides a cycle at the speed of 15 km/h but stops for 10 minute to take rest every 20 kms. How much time will he take to cover a distance of 150 kms?
    (a) 11 hours 10 minutes
    (b) 11 hours
    (c) 12 hours 30 m
    (d) 15 hours
17. The ratio between the speed of bike and train is 15 : 27 respectively. Also a bus cover a distance of 720 km in 9 hours. The speed of the bike is three fourth the speed of bus. How much distance will the train cover in 7 hours?
    (a) 756 km
    (b) 760 km
    (c) 740 km
    (d) Cannot be determined
    (e) None of these
18. A bus started its journey from Bettiah and reached Motihari in 44 minute with its average speed 50 km/h. If the average speed of bus is increased by 5 km/h how much time will it take to cover the same distance?
    (a) 31 min
    (b) 36 min
    (c) 38 min
    (d) 40 min
    (e) None of these
19. A train can travel 50% faster than a bus. Both start from point A at the same time and reach point B at same time. Point B is 75 km away from point A on the way the train stopping 12.5 minutes. The speed of bus is:

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(a) 110 km/h
(b) 120 km/h
(c) 55 km/h
(d) 60 km/h
(e) None of these

20. A train crosses a pole in 10 seconds and a platform, which is 40% longer than the length of train in 24 seconds. If the length of platform is 140 m, what is speed of the train?
(a) 36 m/s
(b) 5 m/s
(c) 10 m/s
(d) 15 m/s
(e) None of these

21. It takes eight hours for a 600 km journey if 120 km is done by train and rest by car. It takes 20 minutes more if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of car is: (a) 2 : 3
(b) 3 : 2
(c) 4 : 3
(d) 3 : 4
(e) None of thes

22. Aditya is travelling on his cycle and has calculated that he will reach point A at 2 pm, if he travel at 10 km/h. He will reach there at 12 noon, if he travel at 15 km/h. At what speed must he travel to reach D at 1 pm?
(a) 10 kmph
(b) 11 kmph
(c) 12 kmph
(d) 13 kmph
(e) None of these

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23. A train running at a speed of 40 m/s crosses a pole in 21 seconds less than the time it required to cross a bridge 3.5 times its length at the same speed. What is the length of the bridge?
(a) 1080 m
(b) 240 m
(c) 840 m
(d) 560 m
(e) None of these

24. Two trains A and B start at the same time from Delhi and Patna respectively towards each other they met after 16 hours. If the distance between Delhi and Patna is 1872 km and train B runs 27 km/h faster than train A what is the speed of train A?
(a) 45 km/h
(b) 72 km/h
(c) 48 km/h
(d) 60 km/h
(e) None of these

25. Two trains M and N crosses a pole in 25 seconds and 1 min 15 seconds respectively. Length of train M is half length of train N. What is the respective ratio between speed of train M and train N?
(a) 2 : 3
(b) 1 : 2
(c) 2 : 1
(d) 3 : 2
(e) None of these

26. A truck cover a distance of 396 km in 5 hour 30 minutes. The average speed of bike is 4 3 of average speed of truck. What is the time taken by bike to cover distance of 12 km less than the distance which the truck cover?

(a) 4 Hrs
(b) 3 Hrs
(c) 5 Hrs
(d) 4 hour 15 minutes
(e) None of these

27. Train A starts its journey from Patna to Hazipur while train B starts from Hazipur to Patna. After crossing each other they finish their journey in 81 hours and 121 hours respectively. Then what will be speed of train B if train A speed is 44 km/h?
(a) 44 Km/h
(b) 55 Km/h
(c) 36 Km/h
(d) 46 Km/h
(e) None of these

28. Two train start at same time from Delhi and Mumbai and comes towards each other at the rate of 85 km and 65 km per hour respectively. When they meet it is found that one train has travelled 20 km more than the other. Find the distance between Delhi and Mumbai?
(a) 140 Km
(b) 75 Km
(c) 150 Km
(d) Cannot be determined
(e) None of these

29. A train start from Delhi on Sunday at 9 am with speed of 25 km/h, another train starts from there on same day at 40 km/h at 3 pm in the same direction. Find at what distance from Delhi both train will meet?
(a) 200 Km
(b) 300 Km
(c) 400 Km
(d) Cannot be determined
(e) None of these

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30. An aeroplane covers a certain distance at a speed of 240 km per hour in 5 hours. To cover the same distance in 5/3 hours, it must travel at a speed of :
(a) 300 km./hr.
(b) 360 km./hr
(c) 600 km./hr
(d) 720 km./hr
(e) None of these

31. A boy goes to his school from his house at a speed of 3 km/hr. If he takes 5 hours in going and coming, the distance between his house and school is:
(a) 6 km
(b) 5 km
(c) 5.5 km
(d) 7.5 km
(e) None of these

32. If a train runs at 40 km/hour, it reaches its destination late by 11 minutes. But if it runs at 50 km/hour, it is late by 5 minutes only. The correct time (in minutes) for the train to complete the journey is
(a) 13
(b) 15
(c) 19
(d) 21
(e) None of these

33. A train 800 meter long is running at the speed of 78 km/hr. if it crosses a tunnel in 1 minute, then the length of the tunnel (in meter) is :
(a) 772

(b) 500
(c) 1300
(d) 13
(e) None of these

34. A cyclist, after cycling a distance of 70 km on the second day, finds that the ratio of distance covered by him on the first two days is 4 : 5. If he travels a distance of 42 km. on the third day, then the ratio of distances travelled on the third day and the first day is:
(a) 3 : 4
(b) 5 : 2
(c) 9 : 4
(d) 9 : 13
(e) None of these

35. A thief is noticed by a policeman from a distance of 200m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km. and 11 km. per hour respectively. What is the distance between them after 6 minutes?
(a) 100 m
(b) 190 m
(c) 200 m
(d) 150 m
(e) None of these

36. A boy started from his house by bicycle at 10 a.m. at a speed of 12 km per hour. His elder brother started after 1 hr 15 mins by scooter along the same path and caught him at 1.30 p.m. The speed of the scooter will be (in km/hr):
(a) 4.5
(b) 36
(c) 56/3
(d) 9
(e) None of these

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37. A student walks from his house at a speed of 2 km per hour and reaches his school 6 minutes late. The next day he increases his speed by 1 km per hour and reaches 6 minutes before school time. How far is the school from his house?
(a) 5/4 km
(b) 6/5 km
(c) 9/4 km
(d) 11/4 km
(e) None of these

38. A moving train crosses a man standing on a platform and a bridge of 300 metres long in 10 seconds and 25 seconds respectively. What will be the time taken by the train to cross a platform 200 metres long? (a) 50/3 seconds
(b) 18 seconds
(c) 20 seconds
(d) 22 seconds
(e) None of these

39. A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and C in 198 seconds, all starting at the same point. After what time will they next meet at the starting point again?
(a) 46 minutes 12 seconds
(b) 45 minutes
(c) 42 minutes 36 seconds
(d) 26 minutes 18 seconds
(e) None of these

40. A and B run a kilometer and A wins by 25 sec. A and C run a kilometer and A wins by 275 m. When B and C run the same distance, B wins by 30 sec. The time taken by A to run a kilometer is
    (a) 2 min 25 sec.
    (b) 2 min 50 sec.
    (c) 3 min 20 sec.
    (d) 3 min 30 sec.
    (e) None of these

Level -3 Difficult

1. Aditya covered a certain distance at some speed. Had he moved 3 km/h faster he would have taken 40 minute less. If he had moved 2 km/h slower he would have taken 40 minutes more. The distance is:
    (a) 40
    (b) 35
    (d) 45
    (c) 49
    (e) None of these
2. A bus take 15 hours to go from Delhi to Chandigarh. Bus go one-third time with speed of 80 km/h. 40% of rest time with speed of 70 km/h. 2/3rd of rest of time with speed of 85 km/h rest with speed of 100 km/ h. What is the distance that cover by bus?
    (a) 1200
    (b) 860
    (d) 1220
    (c) 1120
    (e) None of these
3. In a race of one kilometer, A gives B a start of 100 metres and still wins by 20 seconds. But if gives B a start of 25 seconds, B wins by 50 metres. The time taken by A to run one kilometer is:
    (a) 500/29 seconds
    (d) 700/29 seconds
    (b) 1200/29 seconds
    (e) None of these
4. A man covers a certain distance between his house and office on scooter. Having an average speed of 30 km/hr, he is late by 10 min. However, with a speed of 40 km/hr, he reaches his office 5 min earlier. Find the distance between his house and office.
    (a) 20 km
    (d) 35 km
    (b) 25 km
    (c) 30 km
    (e) None of these
5. The distance between two stations, Delhi and Amritsar, is 450 km. A train starts at 4 p.m. from Delhi and moves towards Amritsar at an average speed of 60 km/hr. Another train starts from Amritsar at 3.20 p.m. and moves towards Delhi at an average speed of 80 km/hr. How far from Delhi will the two trains meet and at what time?
    (a) 5.30 p.m.
    (d) 6.30 a.m.
    (b) 5.50 a.m.
    (c) 6.50 p.m.
    (e) None of these
6. A, B and C can walk at the rates of 3, 4 and 5 km per hour respectively. They starts from Pune at 1, 2, 3 o'clock respectively. When B catches A, B sends him back with a message to C. When will C get the message?

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(a) 4:15 o'clock
(b) 5:15 o'clock
(c) 6:25 o'clock
(d) Can't be determined
(e) None of these

7. Two trains start at the same time from two stations and proceed towards each other at the rates of 20 km and 25 km per hour respectively. When they meet, it is found that one train has travelled 80 km more than the other. Find the distance between the two stations.
   (a) 720 km
   (b) 740 km
   (d) 780 km
   (c) 760 km
   (e) None of these
8. A goods train and a passenger train are running on parallel tracks in the same direction. The driver of the goods train observes that the passenger train coming from behind overtakes and crossed his train completely in 60 seconds. Whereas a passenger on the passenger train marks that he crossed the goods train in 40 seconds. If the speed of the train be in the ratio of 1 : 2, find the ratio of their lengths.
   (a) 1 : 2
   (b) 2 : 1
   (d) 2 : 3
   (c) 3 : 2
   (e) None of these
9. A train after travelling 50 km meets with an accident and then proceeds at 3 4 of its former speed and arrives at its destination 35 minutes late. Had the accident occurred 72 km further, it would have reached the destination only 15 minutes late. The normal speed of the train is
   (a) 36 km/hr.
   (b) 38 km/hr.
   (d) 72 km/hr.
   (c) 46 km/hr.
   (e) None of these
10. Two trains measuring 100 m and 80 m respectively, run on parallel lines of track. When travelling in opposite directions they are observed to pass each other in 9 seconds, but when they are running in the same direction at the same rate as before, the faster train passed the other in 18 seconds. Find the speed of the two trains in km per hour.
    (a) 12 km/hr, 5 km/hr
    (b) 14 km/hr, 18 km/hr
    (c) 16 km/hr, 54 km/hr
    (d) 18 km/hr, 54 km/hr
    (e) None of these
11. A 300 meters long train is travelling with the speed of 45 km/hr when it passes point A completely. At the same time, a motorbike starts from point A with the speed of 70 km/hr. When it exactly reaches the middle point of the train, the train increases its speed to 60 km/hr and motorbike reduces its speed to 65 km/hr. How much distance will the motorbike travel while passing the train completely?
    (a) 3.8 km
    (b) 3.2 km
    (d) 2.2 km
    (c) 2.37 km
    (e) None of these
12. Two trains, Kanpur Mail and Delhi Mail, start at the same time from stations Kanpur and Delhi respectively towards each other. After passing each other, they take 12 hours and 3 hours to reach Delhi and Kanpur respectively. If the Kanpur Mail is moving at the speed of 48 km/hr, the speed of the Delhi Mail is

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(a) 90 km/hr
(b) 96 km/hr
(d) 84 km/hr
(c) 86 km/hr
(e) None of these

13. Heena and Renu are competing in a 100 meters race. Initially, Heena runs at twice the speed of Renu for the first fifty meters. After the 50 meters mark, Heena runs at 1 th 4 her initial speed while Renu continues to run at her original speed. If Renu catches up with Heena at a distance of N meters from the finish line, then N is equal to:
(a) 75 meters
(b) 55 meters
(d) 30 meters
(c) 25 meters
(e) None of these

14. A horse rider travels on horse back from Bhopal to Chandigarh at a constant speed. If the horse increased its speed by 6 km/hr, it would take the rider 4 hours less to cover that distance. And travelling with a speed 6 km/hr lower than the initial speed, it would take him 10 hours more than the time he would have taken if he had travelled at a speed 6 km/hr higher than the initial speed. Find the distance between Bhopal and Chandigarh?
(a) 720 km
(d) 480 km
(b) 680 km
(c) 560 km
(e) None of these

15. Seeta and Geeta cycled towards each other, one from point A and the other from point B, respectively. Seeta left point A 6 hours later than Geeta left point B, and it turned out on their meeting that Seeta had travelled 12 km less than Geeta. After their meeting, they kept cycling with the same speed, and Seeta arrived at B 8 hours later and Geeta arrived at A 9 hours later. Find the speed of the faster cyclist.
(a) 3 km/hr
(d) 12 km/hr
(b) 6 km/hr
(c) 9 km/hr
(e) None of these

16. Two cyclists started simultaneously towards each other and meet each other 3 hours 20 minutes later. How much time will it take the slower cyclist to cover the whole distance if the first arrived at the place of departure of the second 5 hours later than the second arrived at the point of departure of the first? (a) 5 hours
(b) 10 hours
(d) 20 hours
(c) 15 hours
(e) None of these

17. A dog finds a cat at 25 leaps away. The cat sees the dog coming towards it and starts running with the dog in hot pursuit. In every minute, the dog makes 5 leaps and the cat makes 6 leaps and one leap of the dog is equal to 2 leaps of the cat. Find the time in which the dog will catch the cat.
(a) 12.5 minutes
(d) 10.5 minutes
(b) 13 minutes
(c) 11.5 minutes
(e) None of these

18. Two guns were fired from the same place at an interval of 13 minutes but a person in a train approaching the place hears the second shot 12 minutes 30 seconds after the first. Find the speed of the train, supposing that sound travels at 330 metres per second.

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(a) 1188/25 km/hr
(b) 1186/25 km/hr
(c) 1184/25 km/hr
(d) 1187/25 km/hr
(e) None of these

19. A man walks from A to B and back in a certain time at the rate of 7/2 km per hour. But if he had walked from A to B at the rate of 3 km an hour and back from B to A at the rate of 4 km an hour, he would have taken 5 minutes longer. Find the distance between A and B.
(a) 14 km
(b) 12 km
(d) 7 km
(c) 6 km
(e) None of these

20. A carriage driving in a fog passed a man who was walking at the rate of 3 km an hour in the same direction. He could see the carriage for 4 minutes and it was visible to him upto a distance of 100m. What was the speed of the carriage?
(a) 11/2
(b) 9/2
(c) 7/2
(d) 17/2
(e) None of these

21. A man leaves a point P and reaches the point Q in 4 hours. Another man leaves the point Q, 2 hours earlier and reaches the point P in 4 hours. Find the time in which the first man meets to the second man.
(a) 2 hours
(b) 3.5 hours
(d) 3 hours
(c) 1 hours
(e) None of these

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22. A man travels 360 km in 4 hrs, partly by air and partly by train. If he had travelled all the way by air, he would have saved 4 5 of the time he was in train and would have arrived at his destination 2 hours early. Find the distance he travelled by train?
(a) 75 km
(d) 80 km
(b) 90 km
(c) 85 km
(e) None of these

23. One aeroplane started 30 minutes later than the scheduled time from a place 1500 km away from its destination. To reach the destination at the scheduled time the pilot had to increase the speed by 250 km/hr. What was the speed of the aeroplane per hour during the journey?
(a) 750 km/hr
(d) 745 km/hr
(b) 755 km/hr
(c) 760 km/hr
(e) None of thes

24. A train leaves the station 1 hour before the scheduled time. The driver decreases its speed by 4 km/hr. At the next station 120 km away, the train reached on time. Find the original speed of the train. (a) 24 km/hr.
(d) 27 km/hr.
(b) 25 km/hr.
(c) 26 km/hr.
(e) None of these

25. A hare sees a dog 100 metres away from her and run off in the opposite direction at a speed of 12 km an hour. A minute later the dog perceives her and gives chase at a speed of 16 km per hour. How much distance cover by hare after being spotted?
    (a) 750 metres
    (d) 900 metres
    (b) 800 metres
    (e) None of these

ANSWER KEY

Level 1 – Foundation

1	c	6	c	11	a	16	b	21	b	26	a	31	a	36	d
2	c	7	b	12	b	17	b	22	c	27	c	32	a	37	c
3	b	8	c	13	b	18	b	23	c	28	a	33	d	38	a
4	d	9	c	14	d	19	a	24	a	29	c	34	c	39	b
5	d	10	c	15	c	20	a	25	c	30	a	35	d	40	c

Level 2 – Moderate

1	b	6	b	11	d	16	a	21	d	26	a	31	d	36	c
2	a	7	c	12	a	17	a	22	c	27	c	32	c	37	b
3	b	8	c	13	b	18	d	23	c	28	c	33	b	38	c
4	c	9	b	14	c	19	b	24	a	29	c	34	a	39	a
5	b	10	b	15	c	20	c	25	d	30	d	35	a	40	c

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Level 3 – Difficult

1	a	6	b	11	c	16	b	21	c
2	d	7	a	12	b	17	a	22	b
3	a	8	b	13	c	18	a	23	a
4	c	9	d	14	a	19	d	24	a
5	c	10	d	15	b	2	b	25	d